WORKED SOLUTION

STEP III 1993, Question 4

Pure • Summing Series

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s3-q4

The key idea

Each series is a familiar one in disguise. The usual way to sum a series is to recognise it as a standard expansion (logarithm, geometric series or its derivative) evaluated at a particular point, or to split the general term so that the sum telescopes. So for each part we ask which standard series it could be, and what we would have to multiply, differentiate or rearrange to get it into that form.

Solution

Part (i)

The odd denominators $1, 3, 5, \ldots$ suggest the logarithm series with only odd powers left. For $|x| < 1$,

$$\ln(1+x) = x - \tfrac{x^2}{2} + \tfrac{x^3}{3} - \cdots, \qquad \ln(1-x) = -x - \tfrac{x^2}{2} - \tfrac{x^3}{3} - \cdots,$$

and subtracting removes the even powers:

$$\ln\frac{1+x}{1-x} = 2\left(x + \frac{x^3}{3} + \frac{x^5}{5} + \cdots\right) = 2\sum_{n=0}^{\infty}\frac{x^{2n+1}}{2n+1}.$$

Our series is $\sum x^{2n}/(2n+1)$ with $x = \frac{1}{2}$. The powers are even, so we divide by x (allowed, since $x \neq 0$):

$$\sum_{n=0}^{\infty}\frac{x^{2n}}{2n+1} = \frac{1}{2x}\ln\frac{1+x}{1-x}.$$

At $x = \frac{1}{2}$, which is inside the interval of convergence, $\frac{1+x}{1-x} = \frac{3/2}{1/2} = 3$ and $\frac{1}{2x} = 1$, so

$$1 + \frac{1}{3}\left(\frac{1}{2}\right)^2 + \frac{1}{5}\left(\frac{1}{2}\right)^4 + \cdots = \ln 3.$$

Part (ii)

The coefficients repeat with period 4: they are $2, -1, 0, -1$ for $x^{4k}, x^{4k+1}, x^{4k+2}, x^{4k+3}$. So we group each block of four terms:

$$\sum_{k=0}^{\infty} x^{4k}(2 - x - x^3) = \frac{2 - x - x^3}{1 - x^4},$$

a geometric series with ratio x^4, which converges since $|x^4| < 1$. The grouping is legitimate because the original series converges absolutely for $|x| < 1$: its terms are bounded in size by $2|x|^m$.

Now simplify. The numerator vanishes at $x = 1$, so $1 - x$ is a factor:

$$2 - x - x^3 = (1 - x)(x^2 + x + 2), \qquad 1 - x^4 = (1 - x)(1 + x)(1 + x^2).$$

Cancelling $1 - x$ (non-zero, as $|x| < 1$),

$$\frac{x^2 + x + 2}{(1 + x)(1 + x^2)} = \frac{A}{1 + x} + \frac{Bx + C}{1 + x^2}.$$

Putting $x = -1$ gives $2 = 2A$, so $A = 1$. Then $x^2 + x + 2 - (1 + x^2) = x + 1 = (Bx + C)(1 + x)$, so $B = 0$ and $C = 1$. Therefore

$$2 - x - x^3 + 2x^4 - \cdots = \frac{1}{1 + x} + \frac{1}{1 + x^2}.$$

This is easy to confirm: $1 - x + x^2 - x^3 + x^4 - \cdots$ plus $1 - x^2 + x^4 - \cdots$ gives coefficients $2, -1, 0, -1, 2, \ldots$, as required.

Part (iii)

Write the general term as

$$\frac{r\,2^{r-2}}{3^{r-1}} = \frac{1}{2}\, r \left(\frac{2}{3}\right)^{r-1}.$$

Ask yourself Where have we seen $\sum r y^{r-1}$ before?

It is the derivative of the geometric series. For $|y| < 1$, differentiating $\sum_{r=0}^{\infty} y^r = \frac{1}{1-y}$ term by term (valid inside the interval of convergence) gives

$$\sum_{r=1}^{\infty} r y^{r-1} = \frac{1}{(1 - y)^2}.$$

With $y = \frac{2}{3}$ this is $\frac{1}{(1/3)^2} = 9$. Our sum starts at $r = 2$, so we remove the $r = 1$ term, which is 1:

$$\sum_{r=2}^{\infty} \frac{r\,2^{r-2}}{3^{r-1}} = \frac{1}{2}(9 - 1) = 4.$$

Part (iv)

Since $r(r^2-1) = (r-1)r(r+1)$ is a product of three consecutive integers, we look for a difference of two consecutive "pair" fractions:

$$\frac{1}{(r-1)r} - \frac{1}{r(r+1)} = \frac{(r+1)-(r-1)}{(r-1)r(r+1)} = \frac{2}{r(r^2-1)}.$$

So the sum telescopes. Summing from $r = 2$ to N, every term cancels except the first and last:

$$\sum_{r=2}^{N} \frac{2}{r(r^2-1)} = \frac{1}{1\cdot 2} - \frac{1}{N(N+1)}.$$

As $N \to \infty$ the last term tends to 0, so

$$\sum_{r=2}^{\infty} \frac{2}{r(r^2-1)} = \frac{1}{2}.$$

Checks

- (i): the first four terms give $1 + 0.0833 + 0.0125 + 0.0022 \approx 1.098$, already close to $\ln 3 \approx 1.0986$.
- (ii): at $x = 0$ both sides equal 2. At $x = \frac{1}{2}$ the closed form gives $\frac{2}{3} + \frac{4}{5} = \frac{22}{15}$, which matches the partial sums numerically.
- (iii): the first terms are $\frac{2}{3}, \frac{2}{3}, \frac{16}{27}, \ldots$; partial sums climb slowly towards 4.
- (iv): the partial sum to $N = 4$ is $\frac{1}{3} + \frac{1}{12} + \frac{1}{30} = \frac{9}{20} = \frac{1}{2} - \frac{1}{20}$, matching the formula.

Common slips

- In (i), forgetting to divide by x: $\sum x^{2n+1}/(2n+1)$ at $x = \frac{1}{2}$ is $\frac{1}{2}\ln 3$, not $\ln 3$.
- In (iii), summing from $r = 1$ and forgetting to subtract the missing first term, which gives $\frac{9}{2}$.
- In (iv), telescoping with $\frac{1}{r-1} - \frac{1}{r+1}$ instead: that splits $\frac{2}{r^2-1}$, not $\frac{2}{r(r^2-1)}$.

Beyond STEP

The steps we took without fuss (regrouping terms in (ii) and differentiating a power series term by term in (iii)) are exactly what a first analysis course justifies. Inside its radius of convergence a power series converges absolutely and can be differentiated term by term, but rearranging a series that converges only conditionally can change its sum (Riemann's rearrangement theorem). Abbott's *Understanding Analysis* covers both carefully.

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